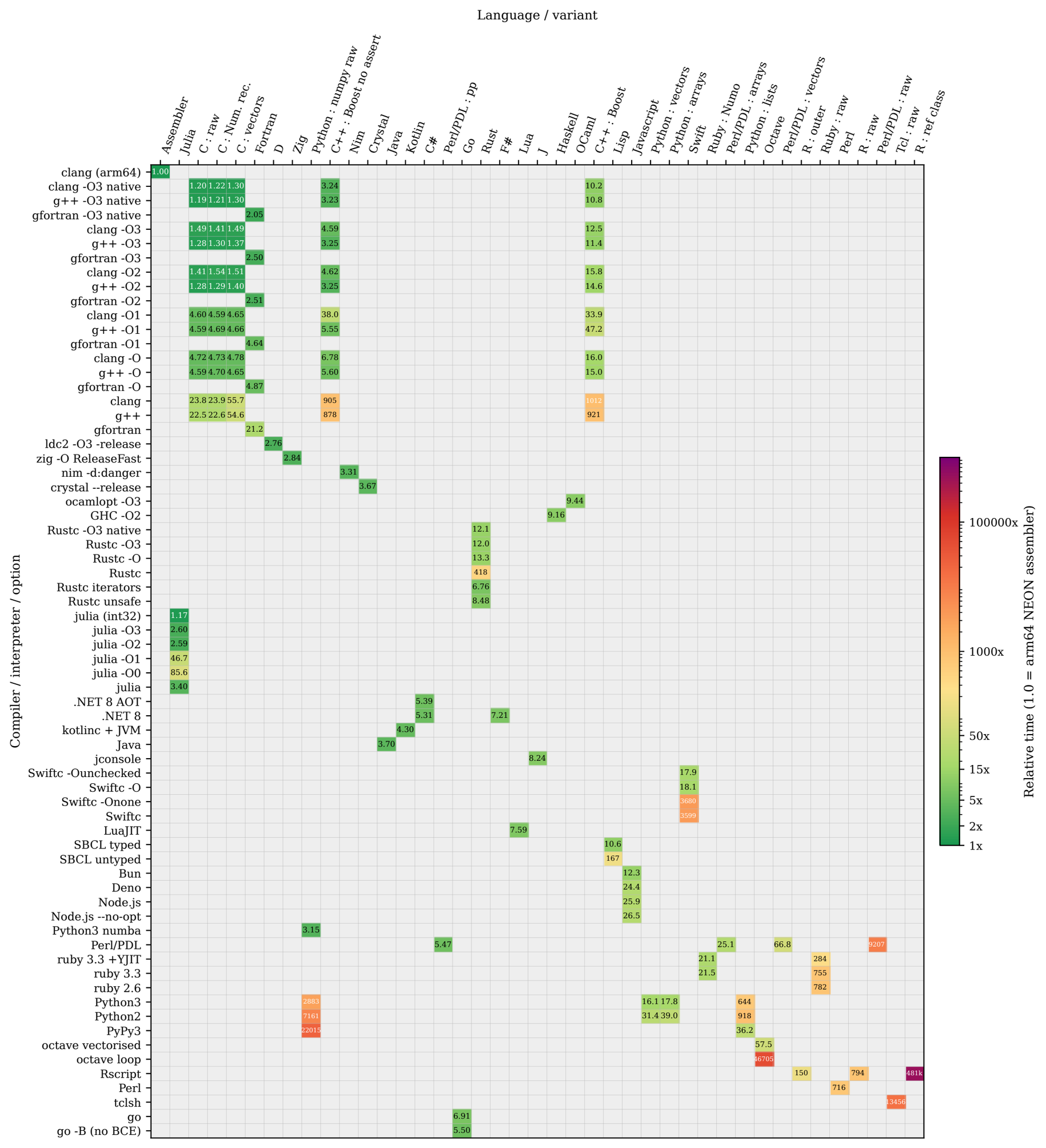
Language / variant
Assembler
Julia
C : raw
C : Num. rec.
C : vectors
Fortran
D
Zig
Python : numpy raw
C++ : Boost no assert
Nim
Crystal
Java
Kotlin
C#
Perl/PDL : pp
Go
Rust
F#
Lua
J
Haskell
OCaml
C++ : Boost
Lisp
Javascript
Python : vectors
Python : arrays
Swift
Ruby : Numo
Perl/PDL : arrays
Python : lists
Octave
Perl/PDL : vectors
R : outer
Ruby : raw
Perl
R : raw
Perl/PDL : raw
Tcl : raw
R : ref class
Compiler / interpreter / option
clang (arm64)
clang -O3 native
g++ -O3 native
gfortran -O3 native
clang -O3
g++ -O3
gfortran -O3
clang -O2
g++ -O2
gfortran -O2
clang -O1
g++ -O1
gfortran -O1
clang -O
g++ -O
gfortran -O
clang
g++
gfortran
ldc2 -O3 -release
zig -O ReleaseFast
nim -d:danger
crystal --release
ocamlopt -O3
GHC -O2
Rustc -O3 native
Rustc -O3
Rustc -O
Rustc
Rustc iterators
Rustc unsafe
julia (int32)
julia -O3
julia -O2
julia -O1
julia -O0
julia
.NET 8 AOT
.NET 8
kotlinc + JVM
Java
jconsole
Swiftc -Ounchecked
Swiftc -O
Swiftc -Onone
Swiftc
LuaJIT
SBCL typed
SBCL untyped
Bun
Deno
Node.js
Node.js --no-opt
Python3 numba
Perl/PDL
ruby 3.3 +YJIT
ruby 3.3
ruby 2.6
Python3
Python2
PyPy3
octave vectorised
octave loop
Rscript
Perl
tclsh
go
go -B (no BCE)
Relative time (1.0 = arm64 NEON assembler)
100000x
1000x
50x
15x
5x
2x
1x